Vinay Prakash

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bitbucket.org/vinaypra/workspace/repositories/

EXPERIENCE

Electrical Integration Engineer (Autonomy Team), Monarch Tractor *Nov 2022 - Present*
- Maintained and refactored the SW-FW interface layer for 4 CAN buses on NVIDIA Orin CAN interfaces for autonomy.
- Devised CAN interface bringup, diagnostics and error handling SW policies from prototype to production tractors.
- Delivered a metrics and diagnostic pipeline for value-added insights for customer about operations and tractors.
- Executed the HRI lights project for FW and SW development compliant with OSHA regulations for Ag environment.
- Led several internal projects like Front Loader implements and ISOBUS abstraction layer for autonomous operations.

Electrical Engineer (Electrical Engineering Team), Monarch Tractor *Jun 2021 - Oct 2022*
- Managed electrical component sizing, sourcing, design, testing & validation for 5 out of 8 electrical sub-systems.
- Migrated 2D electrical harness schematics from Visio to SW Electrical - created 2D electrical schematics, wiring and component libraries, 3D electrical component libraries, worked with mechanical engineers to route harnesses in 3D.
- Designed custom report formats in SW Electrical for BOM, To-From lists, label and wire/cable cut lists, formboard.
- Trained the team on the SW Electrical and resolved wire harness related design issues for all 8 electrical sub-systems.
- Responsible for HW-SW integration and testing for electrical components from protype to production models.
- Performed root cause analysis and troubleshooting for electrical failures in a fleet of 300+ tractors, ensuring minimal downtime and increased system reliability.

Teaching Assistant, Michigan Technological University *Aug 2020 - Apr 2021*
- TA for EE4250: Modern Communication Systems and EE5365: In-Vehicle Communication for Dr. Aurenice Oliveira, supporting a combined class size of 80+ students.

Research Assistant-Connected & Autonomous Vehicles, Michigan Technological University *Aug 2019 - Jun 2021*
- Project 1 - Novel applications of machine learning algorithms for IEEE 802.11 for wireless sensing.
- Project 2 - Performance evaluation of 5G based communication for V2X non-safety applications for CAVs.
- Project 3 - ECU-to-ECU communication performance evaluation using 5G for V2X safety applications.

Electronics R & D Engineer (Hardware Engineering Team), Lithion Power Pvt. Ltd *Dec 2017 - May 2018*
- Developed data logging solution for real-time battery systems monitoring via Thingspeak and AWS based dashboards.
- Migrated data logger to ARMv8-M in 8 months for production with PCB bring-up, debugging and validation/diagnostics.
- Integrated data logger in BMS for remote diagnostics with superuser control features like geo fencing, remote shutdown.
- Implemented SoC and SoH calculation based real-time drive feedback for drivers and superusers via texts and dashboard.

PROJECTS

Autonomous-AI-Racecar
- Migrating the JetRacer platform to Jetson Orin for enhanced AI processing and autonomous racing capabilities.
- Implemented CAN bus motor control via a custom SocketCAN-ROS interface and PIC18F microcontroller.
- Integrating OAK D-Lite stereo vision camera alongside Raspberry Pi Camera v2 for advanced perception.
- Developing motion planning algorithms (Hybrid A* and TEB) using dual-camera input for autonomous navigation.
- Implementing real-time obstacle detection using DepthAI's Object Tracker with KCF, Kalman filter, and Hungarian algorithm.

EDUCATION

Master of Science, Electrical and Computer Engineering *Aug 2019 - Apr 2021*
Michigan Technological University GPA: 3.5
Bachelor of Technology, Electronics and Communication Engineering *Aug 2013 - Jun 2017*
Rajasthan Technical University Grade: First Division

SKILLS

Applications: SolidWorks Electrical, E3 series, RapidHarness, PCAN View, PiSnoop, MATLAB, Vector CAN tools, IAR Workbench, Keil uVision, NI MultiSim, EAGLE, Wireshark
Communication Protocols: CAN, LIN, OBD-II, SPI, I2C, UART, GSM, ISOBUS
Firmware Development: Embedded C/C++, Real-Time Operating Systems (RTOS), Bootloaders
Testing and Validation: Oscilloscopes, Logic Analyzers, Multimeters, Signal Generators
Microcontrollers and Processors: ATMega, ARM, PIC, STM32, ESP32, Nuvoton, SIMCom, Telit, uBlox
Networking (FW-SW Interfaces): CAN, LIN, ISOBUS, Ethernet, TCP/IP, UDP, MQTT, DDS
Programming Languages: Python, C, C++, Bash
Tools and Frameworks: Git, Jenkins, Docker, Yocto, ROS, ROS 2
Project Management: Agile Methodologies, JIRA, Confluence, Jama, Asana, GitHub